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CIS-25: Programming Using C++
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C++ Midway Project

Description:

This program simulates a free-for-all multiplayer deathmatch. I originally designed it as a fun way to "draw names out of a hat," but eventually, this program will act as the framework for a simple role-playing game.

The program starts by prompting the user to input how many combatants are going to participate in the battle, then the user gives each combatant a name. Upon entering a name, the program assigns that combatant a randomly-generated number of hit points, a strength value, and a weapon.

Once all the names have been entered, the program generates a text file with a play-by-play description of the battle, consisting of funny verbs and body parts. Every turn, each combatant attacks every other combatant with a 50% chance to hit. If an attack hits, the damage done is determined by multiplying the combatant's strength value by their weapon damage. This continues until all but one of the combatants' hit points are less than or equal to zero.

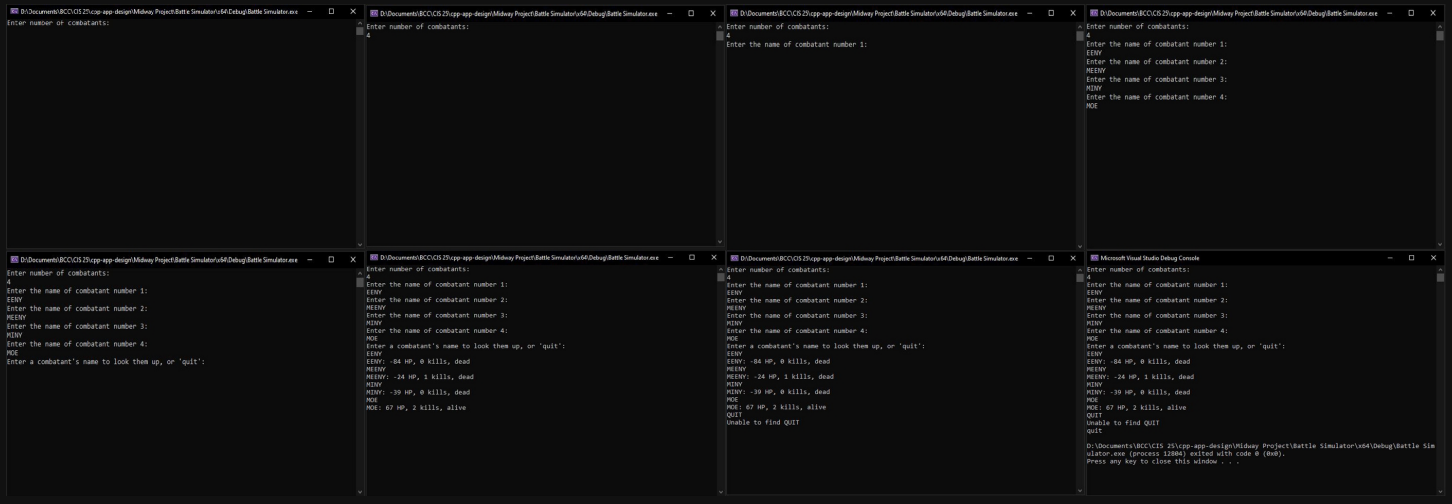
When the battle has concluded, the user is able to search for a name to see how they performed during the battle. This loop continues until the user enters 'quit.'

This program includes or demonstrates all of the requirements for the midway project:

- Data types and data sizes
- File types (e.g., .txt for data I/O)
- Use of **pointers**
- **arrays**
- A **binary search** algorithm (or any other search)
- Use of **strings**
- File I/O (read from or write to a file)
- At least **two classes**

However, even though this program includes multiple inherited classes, only the combatant class is currently functional. The functionality of the other classes will be implemented at a later date.

Output Screenshots:



Output Results:

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1. EENY spawned with 168 HP and 11 strength.
2. MEENY spawned with 108 HP and 19 strength.
3. MINY spawned with 142 HP and 14 strength.
4. MOE spawned with 143 HP and 17 strength.

LET THE GAMES BEGIN!

1. EENY impaled MEENY in the TRACHEA with a a simple KITCHEN KNIFE
MEENY took 22 points of damage!
MEENY's remaining HP: 86
2. EENY tried to attack MINY, but they were distracted by a FLYING PIG!
3. EENY smashed MOE in the STERNUM with a a piece of BOLOGNA
MOE took 11 points of damage!
MOE's remaining HP: 132
4. MEENY mangled EENY in the RIGHT BUTTCHEEK with a a POINTY STICK
EENY took 76 points of damage!
EENY's remaining HP: 92
5. MEENY bludgeoned MINY in the FEMORAL ARTERY with a a pink LIGHT SABER
MINY took 38 points of damage!
MINY's remaining HP: 104
6. MEENY tried to attack MOE, but they were distracted by a FLYING PIG!
7. MINY tried to attack EENY, but they were distracted by a FLYING PIG!
8. MINY tried to attack MEENY, but they were distracted by a FLYING PIG!
9. MINY impaled MOE in the LEFT VENTRICLE with a a TRASH CAN LID
MOE took 14 points of damage!
MOE's remaining HP: 118
10. MOE pelted EENY in the KIDNEY with a a FORK
EENY took 34 points of damage!
EENY's remaining HP: 58
11. MOE smacked MEENY in the KIDNEY with a a FORK
MEENY took 17 points of damage!
MEENY's remaining HP: 69
12. MOE smashed MINY in the CLAVICLE with a a BRICK
MINY took 102 points of damage!
MINY's remaining HP: 2
13. EENY tried to attack MEENY, but they were distracted by a FLYING PIG!
14. EENY impaled MINY in the TRACHEA with a a piece of BOLOGNA
MINY took 11 points of damage!
MINY's remaining HP: -9
------< EENY killed MINY! >------
15. EENY clobbered MOE in the LARGE INTESTINE with a a FORK
MOE took 11 points of damage!
MOE's remaining HP: 107
16. MEENY pelted EENY in the RIGHT BUTTCHEEK with a a rubber DUCKY
EENY took 19 points of damage!
EENY's remaining HP: 39
17. MEENY mangled MOE in the EYEBALL with a a FORK
MOE took 38 points of damage!
MOE's remaining HP: 69
18. MOE tried to attack EENY, but they were distracted by a FLYING PIG!
19. MOE bludgeoned MEENY in the LEFT VENTRICLE with a a pair of POLKDA-DOTTED UNDERWEAR
MEENY took 34 points of damage!
MEENY's remaining HP: 35
20. EENY mangled MEENY in the LUNG with a a simple KITCHEN KNIFE
MEENY took 44 points of damage!
MEENY's remaining HP: -9
------< EENY killed MEENY! >------
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21. EENY smashed MOE in the STERNUM with a a giant BATTLE AXE
MOE took 33 points of damage!
MOE's remaining HP: 36
22. MOE tried to attack EENY, but they were distracted by a FLYING PIG!
23. EENY tried to attack MOE, but they were distracted by a FLYING PIG!
24. MOE tried to attack EENY, but they were distracted by a FLYING PIG!
25. EENY smacked MOE in the MEDULLA OBLONGATA with a a GLADIUS
MOE took 33 points of damage!
MOE's remaining HP: 3
26. MOE stabbed EENY in the LUNG with a a pink LIGHT SABER
EENY took 68 points of damage!
EENY's remaining HP: -29
------< MOE killed EENY! >------
MOE is the last combatant standing!

   SCOREBOARD
1. EENY: 2 kills
2. MOE: 1 kills (SURVIVOR!)
3. MEENY: 0 kills
4. MINY: 0 kills